NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

Faculty Name: Dr. Durga Prasad Mohapatra

Department : Computer Sc. & Engg. NIT Rourkela

TOPIC: Project Monitoring and Control



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

CONCEPTS COVERED

- Introduction
- Project Control Cycle
- Project reporting structures
- Assessing progress



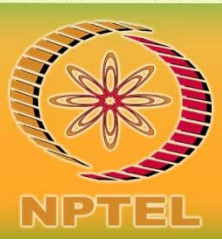
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

NPTEL

Introduction

- Once work schedules have been published and the project is started,
  - attention must be focused on progress.
- This requires
  - monitoring of what is happening,
  - comparison of actual achievement against the schedule and,
  - where necessary, revision of plans and schedules to bring the project as far as possible back on target.

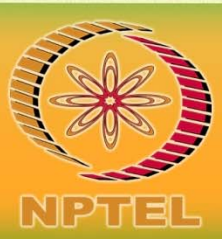




Introduction

- The project manager designates certain key events such as completion of some important activity as a **milestone.**
- **A few examples of milestones are:** preparation and review of the SRS document, completion of the coding and unit testing, etc.

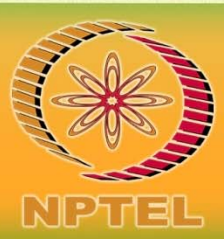




Introduction cont...

- Once a milestone is reached, the project manager can assume that some measurable progress has been made.
- If any delay in reaching a milestone is predicted, then corrective actions might have to be taken.
- This may entail reworking all the schedules and producing a fresh schedule.

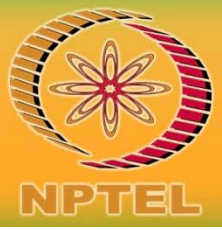




Introduction cont...

- PERT (Project Evaluation and Review Technique) chart is especially useful in project monitoring and control. A path in this graph is any set of consecutive nodes and edges from the starting node to the last node.
- A critical path in this graph is a path along which every milestone is critical to meeting the project deadline. In other words, if any delay occurs along a critical path, the entire project would get delayed.
- It is therefore necessary to identify all the critical paths in a schedule.

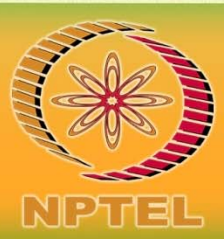




Introduction cont...

- There may be more than one critical path in a schedule. The tasks along a critical path are called critical tasks.
- The critical tasks need to be closely monitored and corrective actions need to be initiated as soon as any delay is noticed.
- If necessary, a project manager may switch resources from a non-critical task to a critical task so that all milestones along the critical path are met.

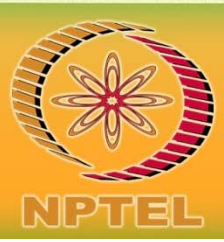




Introduction cont...

- Several tools are available which can help you to figure out the critical paths in an unrestricted schedule, but figuring out an optimal schedule with resource limitations and with a large number of parallel tasks is a very hard problem.
- There are several commercial products for automating the scheduling techniques, are available. Popular tools to help draw the schedule-related graphs include the MS-Project software available on personal computers.

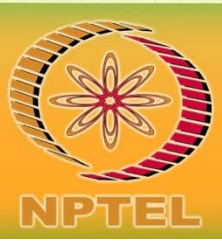




Introduction cont...

- In earlier classes, we have discussed the importance of producing plans that can be monitored.
- Now, we will discuss how information about the progress of the project is gathered and what actions must be taken to ensure that a project meets its targets.

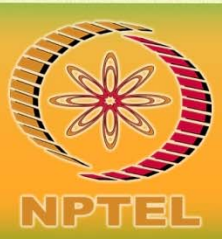




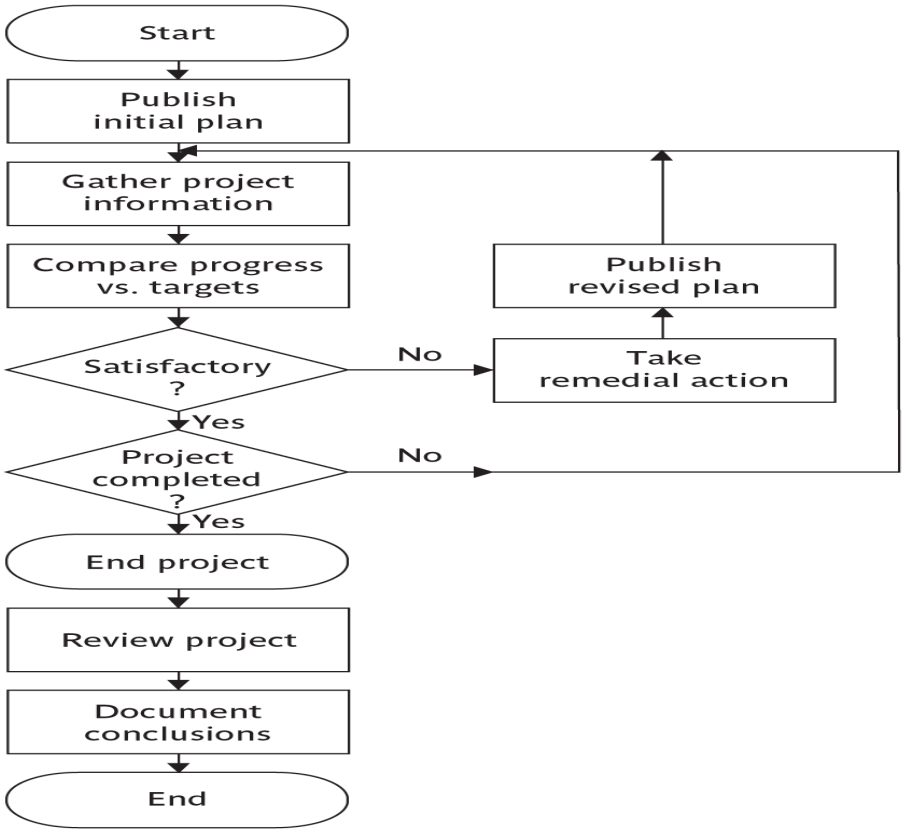
Creating the Framework

- Exercising control over a project and ensuring that targets are met is a matter of regular monitoring
  - Finding out what is happening and comparing it with targets.
- There may be a mismatch between the planned outcomes and the actual ones. Then, replanning may be needed to bring the project back on target.
- Alternatively, the target has to be revised.





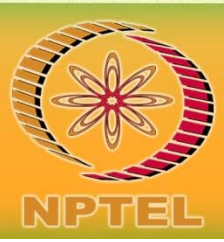
A Model of Project Control Cycle



Creating the Framework cont …

- The project control cycle shows how, once the initial project plan has been published,
  - project control is a continual process of monitoring progress against that plan and, where necessary, revising the plan to take account of deviations.
- It also illustrates the important steps that must be taken after completion of the project so that
  - the experience gained in any one project can feed into the planning stages of future projects,
    - thus allowing us to learn from past mistakes.

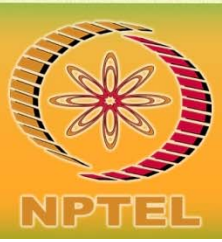




Creating the Framework

- In practice, the project manager normally concerned with four types of shortfall
  - Delays in meeting target dates,
  - shortfalls in quality,
  - inadequate functionality, and
  - costs going over target.

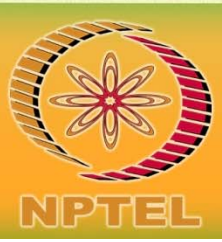




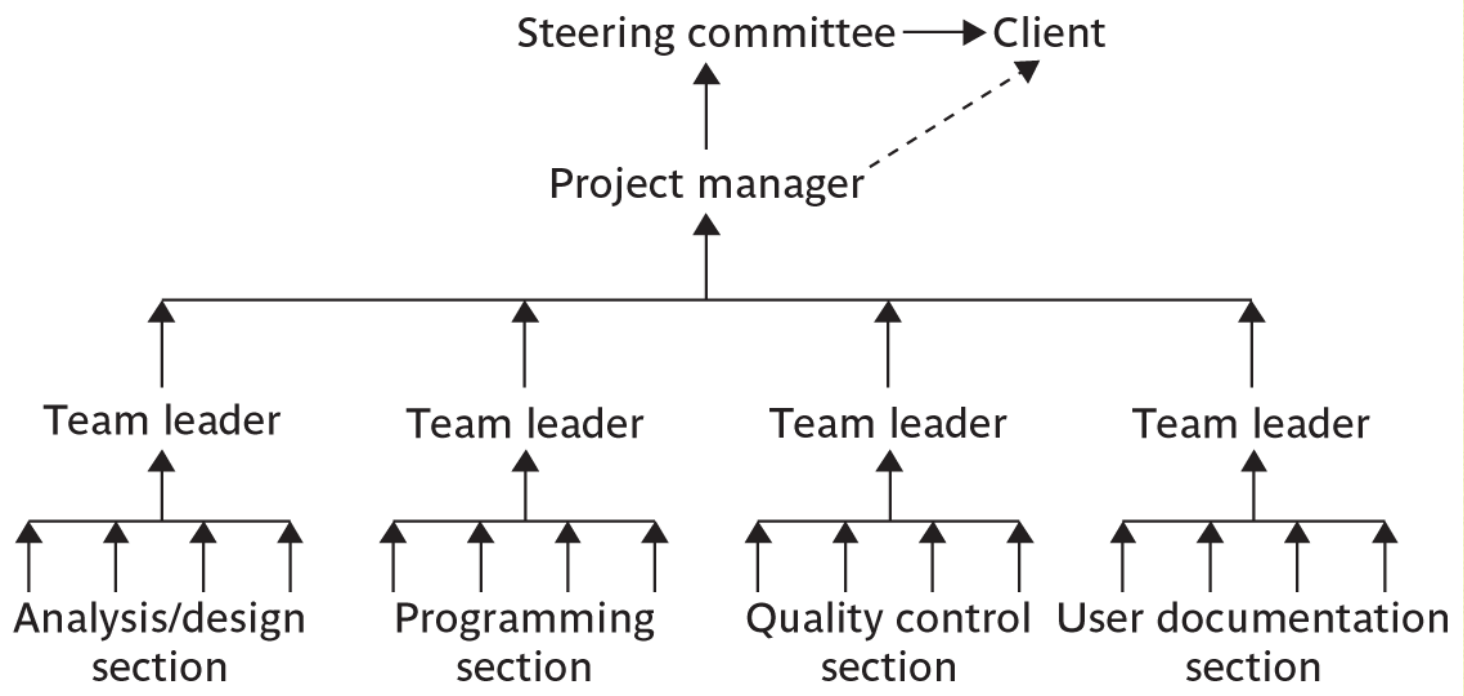
Responsibilities

- The overall responsibility for ensuring satisfactory progress on a project is often the role of the project steering committee, or project management board.
- Day-to-day responsibility will rest with the project manager.
- Other aspects of the projects can be delegated to team leaders.





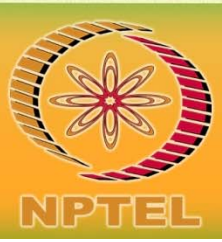
Project reporting structures



Responsibilities cont ...

- The previous figure illustrates the typical reporting structure found with medium and large projects.
- With small projects (employing around half a dozen or fewer staff), individual team members usually report directly to the project manager, but in most cases team leaders will collate reports on their section's progress and forward summaries to the project manager.
- These, in turn, will be incorporated into project-level reports for the steering committee and, via them or directly, progress reports for the client.

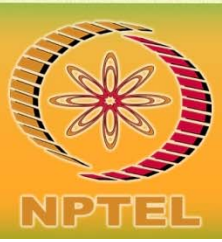




Responsibilities cont ...

- Reporting may be
  - oral or written,
  - formal or informal, and
  - regular or ad hoc.
- Informal communication is necessary and important,
  - but any such informal reporting of project progress must be complemented by formal reporting procedures

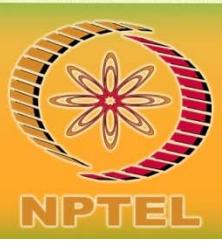




Categories of reporting

Report type	Examples	Comment
Oral formal regular	Weekly or monthly progress meetings	While reports may be oral, formal written minutes should be kept
Oral formal adhoc	End-of-stage review meetings	While largely oral, likely to receive and generate written reports
Written formal regular	Job sheets, progress reports	Normally weekly using forms
Written formal adhoc	Exception reports, change reports	
Oral informal adhoc	Canteen discussion, social interaction	Often provides early warning; must be backed up by formal reporting

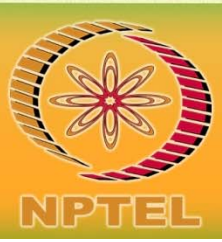




Assessing progress

- Some information used to assess project progress will be collected routinely,
  - while other information will be triggered by specific events.
- Wherever possible, this information should be objective and tangible
  - for example, whether or not a particular report has been delivered





Assessing progress cont ...

- It is essential to set a series of checkpoints in the initial activity plan.

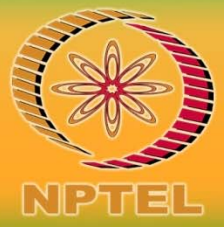
  Checkpoints – predetermined times when progress is checked

  Two types:

  – Event driven: check takes place when a particular event has been achieved (i.e. tied to specific events such as the production of a report or other deliverable.)

  – Time driven: date of the check is pre-determined, or regular (e.g. Weekly / Monthly)

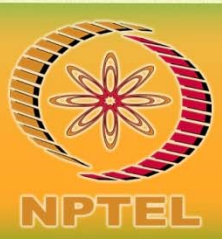




Frequency of reporting

- The frequency of progress reports will depend upon the size and degree of risk of the project.
- The higher the management level, the less frequent (i.e. longer the gaps between checkpoints) and less detailed the reporting needs to be.
- Example: Team leaders may assess progress daily, where as managers do weekly or monthly.
- Major, or project-level, progress reviews generally takes place at particular points during the life of a project
  - commonly known as review points or control points.

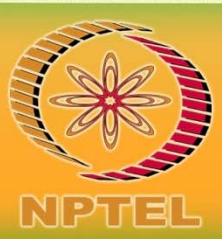




Summary

- Discussed the importance and need of project monitoring and control
- Explained the Project Control Cycle
- Presented the project reporting structures
- Explained how to assess the progress of a project

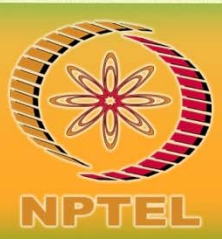




References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.
2. J. Henry, *Software Project Management – A Real-World Guide to Success*, Pearson Education, 2004.
3. R. Mall, *Fundamentals of Software Engineering*, Fifth Edition, PHI Learning Pvt. Ltd., 2018.






Thank you
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
1951
योग कर्मसु कौशलम्
NPTEL